

AJOL download report

ARTICLES

1. Download AJOL

Error downloading metadata from 37 articles (get_html_meta_from_article).

Type of error:

"error reading URL

https://www.ajol.info/index.php/arebusj/article/view/260669 : Error in
read_xml.raw(raw, encoding = encoding, base_url = base_url, as_html =
as_html, : Char 0xFFFF..."

Manually checking the articles all metadata is in html of the web so the reason of the error is unknown.

Articles with error:

- [1] "https://www.ajol.info/index.php/arebusj/article/view/260669"
- [2] "https://www.ajol.info/index.php/ai/article/view/261030"
- [3] "https://www.ajol.info/index.php/ajpherd/article/view/260752"
- [4] "https://www.ajol.info/index.php/ajrh/article/view/260948"
- [5] "https://www.ajol.info/index.php/ajrh/article/view/260908"
- [6] "https://www.ajol.info/index.php/eajmbs/article/view/265411"
- [7] "https://www.ajol.info/index.php/eajmbs/article/view/265412"
- [8] "https://www.ajol.info/index.php/eajmbs/article/view/265413"
- [9] "https://www.ajol.info/index.php/eajmbs/article/view/265414"
- [10] "https://www.ajol.info/index.php/eajmbs/article/view/265456"
- [11] "https://www.ajol.info/index.php/eajhss/article/view/261044"
- [12] "https://www.ajol.info/index.php/eajhss/article/view/261037"
- [13] "https://www.ajol.info/index.php/ijbcs/article/view/265474"
- [14] "https://www.ajol.info/index.php/ijma/article/view/259910"
- [15] "https://www.ajol.info/index.php/jjm/article/view/260529"
- [16] "https://www.ajol.info/index.php/jjm/article/view/260530"
- [17] "https://www.ajol.info/index.php/jjm/article/view/260534"
- [18] "https://www.ajol.info/index.php/jkap/article/view/260684"
- [19] "https://www.ajol.info/index.php/jkap/article/view/260686"
- [20] "https://www.ajol.info/index.php/jkap/article/view/260722"
- [21] "https://www.ajol.info/index.php/mtj/article/view/260896"
- [22] "https://www.ajol.info/index.php/ngjsd/article/view/260705"
- [23] "https://www.ajol.info/index.php/najfnr/article/view/265651"
- [24] "https://www.ajol.info/index.php/najfnr/article/view/265873"
- [25] "https://www.ajol.info/index.php/najfnr/article/view/265874"
- [26] "https://www.ajol.info/index.php/najfnr/article/view/265876"
- [27] "https://www.ajol.info/index.php/najfnr/article/view/265877"
- [28] "https://www.ajol.info/index.php/qm/article/view/260244"

- [29] "<https://www.ajol.info/index.php/qm/article/view/260246>"
- [30] "<https://www.ajol.info/index.php/qm/article/view/260194>"
- [31] "<https://www.ajol.info/index.php/qm/article/view/260198>"
- [32] "<https://www.ajol.info/index.php/qm/article/view/260199>"
- [33] "<https://www.ajol.info/index.php/rjetcs/article/view/261026>"
- [34] "<https://www.ajol.info/index.php/sinet/article/view/259426>"
- [35] "<https://www.ajol.info/index.php/sajg/article/view/265359>"
- [36] "<https://www.ajol.info/index.php/sjms/article/view/245195>"
- [37] "<https://www.ajol.info/index.php/tjpr/article/view/264168>"
- [38] "<https://www.ajol.info/index.php/jnae/article/view/265176>"

Issues with error:

- [1] <https://www.ajol.info/index.php/ajpherd/issue/view/19626>

Articles from these issues were gathered with xml2. Meta was the retrieved with ojsr.

- [1] "<https://www.ajol.info/index.php/bajopas/issue/view/16521/showToc>"
- [2] "<https://www.ajol.info/index.php/jfas/issue/view/16931/showToc>"
- [3] "<https://www.ajol.info/index.php/jfas/issue/view/16929/showToc>"
- [4] "<https://www.ajol.info/index.php/jfas/issue/view/16713/showToc>"
- [5] "<https://www.ajol.info/index.php/pamj/issue/view/17277/showToc>"
- [6] "<https://www.ajol.info/index.php/pamj/issue/view/16223/showToc>"
- [7] "<https://www.ajol.info/index.php/pamj/issue/view/14014/showToc>"
- [8] "<https://www.ajol.info/index.php/pamj/issue/view/13540/showToc>"
- [9] "<https://www.ajol.info/index.php/pamj/issue/view/13578/showToc>"
- [10] "<https://www.ajol.info/index.php/pamj/issue/view/13107/showToc>"
- [11] "<https://www.ajol.info/index.php/pamj/issue/view/12010/showToc>"

2. Metadata analysis

2.1 Metadata fields' population

In total, metadata fields from 237,263 articles were downloaded. Finally, only 223,981 articles were kept as 11,767 missed almost every field. Then, 1,515 duplicates were removed. The fields of the remaining articles were revised to remove false non-NAs (ex. some abstracts' data was 'No abstract'). The final population of each field is displayed in the 'Final N' column. However, some errors might still appear in the data (see following sections for explanation and examples).

Data from the 11,767 articles with NAs (at least from some) can be retrieved scraping the pages of the articles.

Table 1. Metadata fields' population - publications

Field	NAs	% NAs	Selected	Final N
input_url	0	0	1	223981

DC.Creator.PersonalName	11866	5.00	0	-
DC.Date.created	76341	32.18	0	-
DC.Date.dateSubmitted	11767	4.96	0	-
DC.Date.issued	11872	5.00	1	223876
DC.Date.modified	11767	4.96	0	-
DC.Description	11770	4.96	1	192933
DC.Format	17566	7.40	0	-
DC.Identifier	11767	4.96	1	223981
DC.Identifier.pageNumber	69133	29.14	0	-
DC.Identifier.URI	11767	4.96	0	-
DC.Language	11767	4.96	1	-
DC.Rights	11767	4.96	0	-
DC.Source	11767	4.96	1	223981
DC.Source.ISSN	13498	5.70	0	-
DC.Source.Issue	28964	12.21	1	206866
DC.Source.Volume	12298	5.18	1	223450
DC.Source.URI	11767	4.96	0	-
DC.Title	11767	4.96	1	223981
DC.Type	11767	4.96	0	-
DC.Type.articleType	11767	4.96	1	-
gs_meta_revision	11767	4.96	0	-
citation_journal_title	11767	4.96	0	-
citation_journal_abbrev	11767	4.96	0	-
citation_issn	13498	5.70	0	-
citation_author	11785	4.97	0	-
citation_title	11767	4.96	0	-
citation_language	11767	4.96	0	-
citation_date	11767	4.96	0	-
citation_volume	12298	45.18	0	-
citation_issue	28964	12.21	0	-
citation_firstpage	69137	29.14	1	166217
citation_lastpage	69177	29.19	1	166056
citation_abstract_html_url	11767	4.96	0	-
citation_pdf_url	18314	7.72	0	-
DC.Subject	114973	48.46	1	121399
citation_keywords	114973	48.46	0	-
DC.Identifier.DOI	137180	57.82	1	99073
citation_doi	137180	57.82	0	-
citation_author_institution	212079	89.40	0	-
DC.Coverage	237263	100	0	-
citation_reference	237232	99.99	0	-

DC.Title.Alternative	237260	100	0	-
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*Some fields still contain data with errors but cleaning them would require manually revise all data. Examples of erroneous data are provided in the following sections.

**source_url was added to the data from article_url. Then, sourced_id was created to join articles' table to source's tables.

In red, fields that seem to have many wrong values.

2.2 Metadata fields' checks, cleaning and transformations

2.2.1 DOI

A sample of 100 DOIs were randomly selected to check against doi.org.

All of them but one worked (the one that did not work '10.5897/AJB2014.13715' sent me to academicjournals.org but to a 'service temporarily unavailable' page. The DOI appeared in google in academic journals, research gate and AJOL). Most of the DOIs were linked to an AJOL page, but 6 were linked to the journals' page.

2.2.2 Source fields

Source fields were removed. source_url was created from article_url to avoid matching errors between articles' table and source's table (titles written differently, missing ISSNs, etc.). source_url was matched with source's table and substituted by a source_id field (int, 4 digits) to facilitate the matching. All source fields are stored in ajol_source and ajol_source_area.

DC.Source.URI was not used to avoid possible errors.

Note: unique sources in final dataset are 564 because about 176 got removed when the 13,282 articles.

2.2.3 Issue

No changes. Includes numbers and variations of special issues (ex. 'Special', 'S', 'S1', 'SE2').

2.2.4 Volume

No changes. Int.

2.2.5 Date (issued)

There are four dates: created, submitted, issued, modified. No description of the content but created and modified could refer to when the information was uploaded to AJOL, while submitted and issued should refer to submission and issue publication dates.

However, in most cases, both dates are the same.

Manual check of 50 random articles:

All dates in the metadata were the same as the date of the issue (as reported by AJOL). In some cases, the date did not appear in the article webpage, but in the issue webpage. In one case, the publication date of the article website and the issue data

did not match, probably due to a manual coding problem (id 195568, article website reports published 7 May 2020, issue website reports issued 6 May 2020, metadata agrees with issue date).

Year added from issued date.

2.2.6 First and last page

Punctuation (':') was removed. Observations where first and last page were in the same field were splitted (ex. 84-89). Data was transformed to int. Invalid data was removed (ex. '/A', '@yahoo.com').

2.2.7 Type

Unique types were saved in .csv to unify them and create a dictionary. The dictionary is saved as dict_type.csv. References to the same types were unified (ex. 'Original article', 'Article', 'Research article', 'Articles', etc.). Wrong types were classified as NAs (ex. 'Engineering').

Errors have been noticed (ex. a publication classified as 'Article' has 'Book review. Book title: [...]' in the description-abstract).

Quick error check: detected all appearances of 'Book Title' in the abstract field (550) and retrieved the type. Results:

Articles: 533

Book reviews: 9

Research Articles: 5 (then classified as 'Articles' by dict_type.csv):

Review: 3

2.2.8 Title

No changes.

Review of a sample of 500 titles:

- Some titles did not provide information about the content of the publication but the type. Ex: one title is 'Books'; one title is 'Table id content'; one title is 'Letter from the editor'; one title is 'Editorial'; one title is 'Short report'; one title is 'Books reviews'
- The rest of the titles have a title-like format (the webs of the articles were not checked to compare).

2.2.9 Author

Author field was removed from the table.

Authors were separated mainly by semicolons or commas but with a lot of inconsistencies between observations. Commas were also used to divide surname and name of the same author ('Surname, Name'), surname and initials ('Surname, I. I.') or any other combination. The format of the authors was different in many publications (ex. 'Name Surname', 'Name I., Surname', 'I I Surname', etc.). Normalization does not seem possible.

2.2.10 Keywords

Keywords were separated by semicolons, commas or dots. Keywords were divided by their separators and a new table was created for keywords (fields: 'id' and 'keywords').

The terms 'keywords' or 'mesh concepts' were removed.

'Nil' or 'NIL' were removed.

Html tags were removed. Other punctuation symbols (ex. a dot at the end of the field) were also removed.

Did not change to lower letters.

2.2.11 Abstract

A separate table was created for abstracts.

All abstracts with less than 100 characters were removed to remove common terms and sentences for missing abstracts (ex. 'No abstract', 'None', 'All abstracts contained within the PDF.') and citation data (ex. 'SAMJ 45(6): 144-146', 'S. Afr. Med. J., 48, 1495 (1974)').

The word 'Abstract'/'ABSTRACT' at the beginning of the field was removed.

Some abstracts refer to the data of the book they are reviewing.

2.2.12 Language

All articles in the download are coded as English.

Tried to use language identification algorithms to check the languages but does not work. Problems can be due to technical words but also because the abstract is in English and the article in other language, or because the abstract is in two languages and the language of the full-text is unknown.

2. JOURNALS

2.1 Metadata fields' population

Journals' data was scraped from AJOL website. In total, information was retrieved from 739 journals.

Table 2. Metadata fields population - journals

Field	NAs
Title	0
eISSN	84
ISSN print	334
country	0
JPPS status	5
OA	0

2.2 Metadata fields' checks, cleaning and transformations

2.1 ISSNs

ISSNs were retrieved from the about page of the journals.

11 journals miss both eISSN and ISSN print. Those were checked manually in the ISSN portal. Three of them were found and ISSN was added:

Nigerian Journal of Soil Sciences

African journal of health professions education

Nigerian Bar Journal

43 journals report the same eissn and issn_print. Correct medium/issn was retrieved from the ISSN Portal. In 7 cases, it was not possible to identify the correct issn (no records in the ISSN portal).

All ISSNs were checked in the ISSN Portal.

There were 100 ISSNs not recognized by the portal, which reported the following messages:

For eissn:

Awaiting validation record: 48

Unreported record: 22

Provisional record: 4

Cancelled record: 1

For issn_print

Awaiting validation record: 13

Unreported record: 9

Provisional record: 3

2.2 Country

One journal was missing from the country scraper (it was not on the countries' pages). It was manually retrieved from the journal's about page.

2.3 JPPS status

Five titles were missing from the JPPS lists (also missing from the 'all journals' page but appear in the country list). NAs were assigned.

2.4 Research area

Journals can be assigned to a maximum of 3 research areas.

Three journals missing from research area list. NAs were assigned.

2.5 OA

Only Diamond journals appear in the OA page. The rest of the journals were classified as no OA.

Non-OA journals can have OA articles.

AJOL DOIs

Maybe old publications are the ones that do not have DOI? True for all databases but AJOL.

pub_year	Total pubs	No DOI pubs	%
NULL	105	97	92,4
1966	22	22	100,0
1967	18	18	100,0
1968	24	24	100,0
1969	18	18	100,0
1970	22	22	100,0
1971	26	26	100,0
1972	25	25	100,0
1973	24	24	100,0
1974	26	26	100,0
1975	23	23	100,0
1976	23	23	100,0
1977	27	27	100,0
1978	24	24	100,0
1979	26	26	100,0
1980	31	31	100,0
1981	29	29	100,0
1982	26	26	100,0
1984	9	9	100,0
1986	18	18	100,0
1987	83	83	100,0
1988	80	80	100,0
1989	47	47	100,0
1990	55	55	100,0
1991	103	40	38,8
1992	29	29	100,0
1993	7	7	100,0
1994	6	6	100,0
1995	48	14	29,2
1996	56	9	16,1
1997	198	32	16,2
1998	323	46	14,2
1999	740	141	19,1
2000	1353	295	21,8

2001	1981	458	23,1
2002	2220	667	30,0
2003	892	313	35,1
2004	4726	1259	26,6
2005	4617	1602	34,7
2006	5480	1704	31,1
2007	5396	1771	32,8
2008	7770	1963	25,3
2009	8733	2602	29,8
2010	12272	4765	38,8
2011	8940	2944	32,9
2012	10134	7347	72,5
2013	13351	8959	67,1
2014	10629	5945	55,9
2015	14709	8854	60,2
2016	19252	12471	64,8
2017	13730	9033	65,8
2018	15612	10220	65,5
2019	9676	5523	57,1
2020	8485	4785	56,4
2021	12372	8700	70,3
2022	11496	7304	63,5
2023	15536	12213	78,6
2024	1855	1487	80,2

AJOL-Crossref

DOIs registered by AJOL in Crossref start by '10.4314'

AJOL has 82798 DOIs that start with '10.4314'

Crossref has 83145 DOIs registered by AJOL

From those subsets:

- There are 3786 publications from AJOL that are not in Crossref. Most of them from 2023 and 2024

Year	N
2008	2
2009	3
2010	3

2011	2
2013	1
2014	5
2020	36
2022	209
2023	3157
2024	368

- There are 4261 publications from Crossref that do not have a DOI in AJOL. However, they seem to exist in AJOL (random check of 30 publications). Publications without DOI are from different years and journals (from 116 journals, median = 9,5, mean = 36,7). Most common journals:

Journal	Number of absent DOIs
Metamorphosis	99
Journal of Family Ecology and Consumer Sciences /Tydskrif vir Gesinsekologie en Verbruikerswetenskappe	119
Journal Tunisien d'ORL et de Chirurgie Cervico-Faciale	128
Bayero Journal of Pure and Applied Sciences	131
Dutse Journal of Pure and Applied Sciences	201
Tydskrif vir letterkunde	240
Lexikos	258
Tropical Journal of Animal Science	267
South African Journal of Philosophy	295
African Journal for Physical, Health Education, Recreation and Dance	429
Journal de la Recherche Scientifique de l'Universite de Lome	457

Random check of 30 publications:

- Four were in the first download but were one of the 13,282 pubs removed because they had no metadata.
- One not in AJOL